温州肯恩大学
WENZHOU-KEAN UNIVERSITY

GMBA 5885 W03
Strategic Mgt Global Info Sys

Information Systems and Organizations

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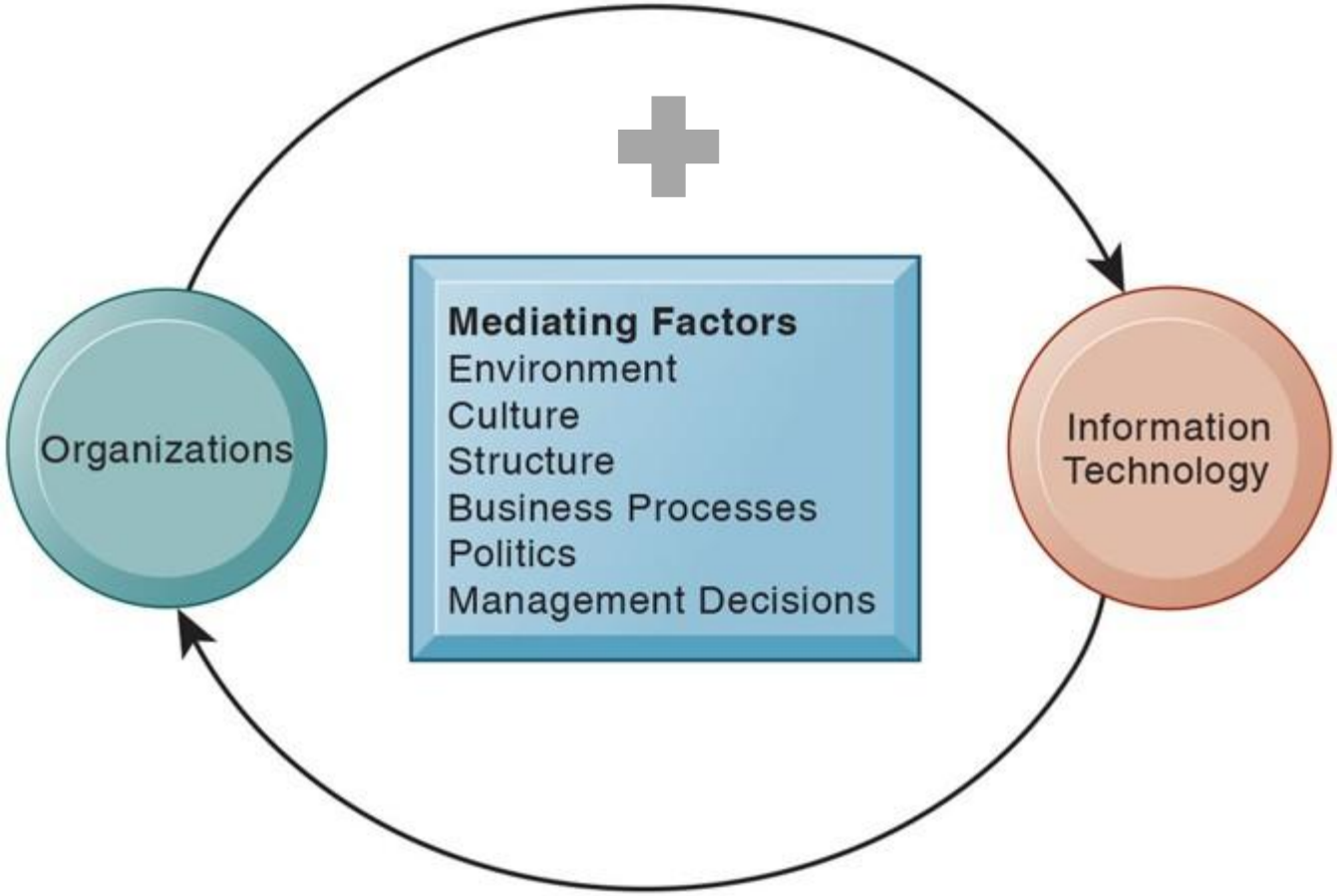
Learning Objectives

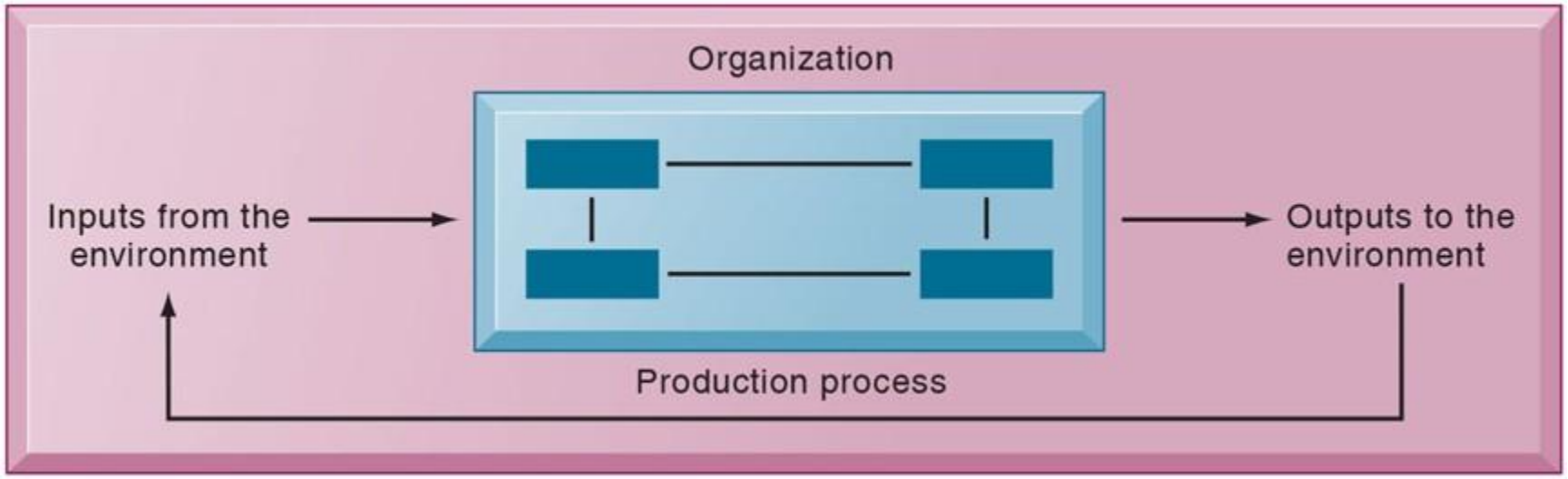
- Understand the technical and behavioral views of organizations.
- Understand how organization and information systems interact to shape business performance and outcomes.
- Identify key organizational features that influence the design and use of information systems.
- Analyze how information systems impact organizational structure, operations, and decision-making.

What would you like to understand about
how organization and information system interact?



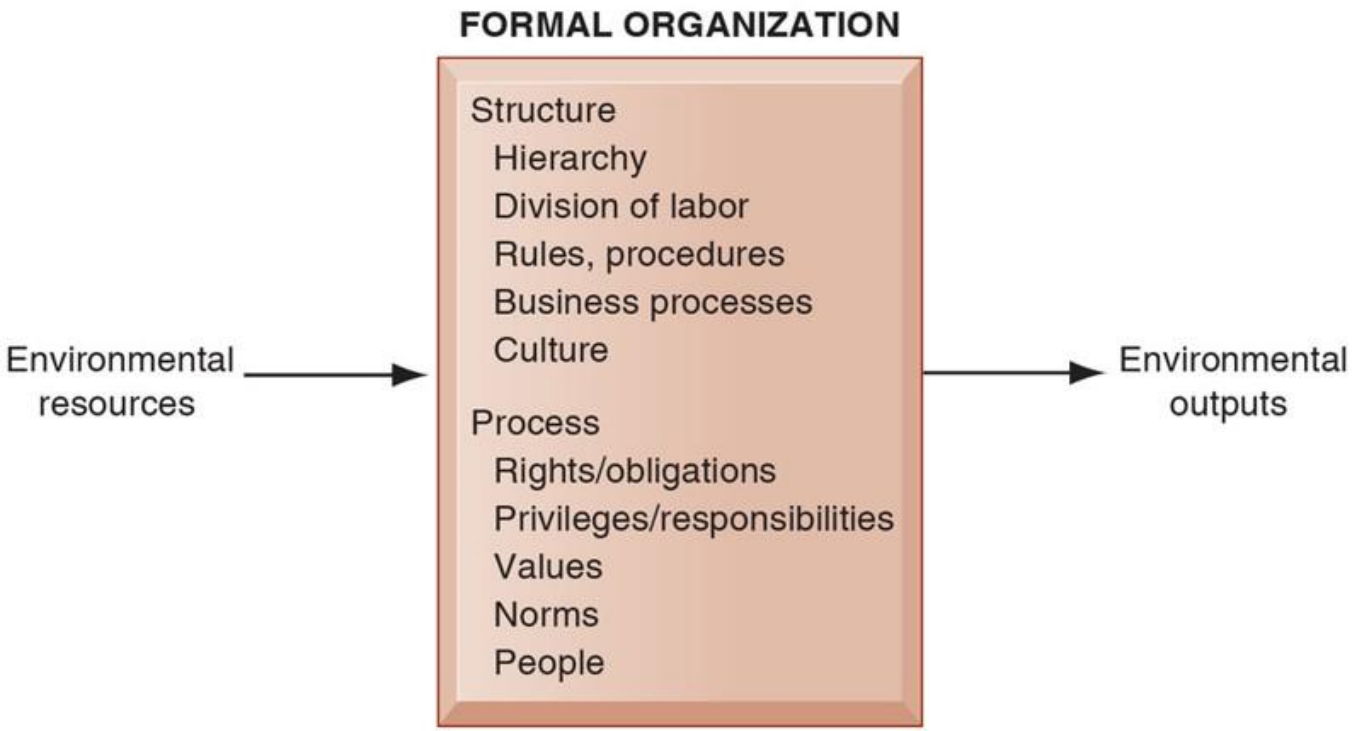
Strategy
Performance

Technical View of Organizations



A formal social structure that transforms **capital** and **labor** into products and services through the production process, which the *environment consumes and supplies* additional capital and labor as input in the <u>feedback loop</u>.

Behavioral View of Organizations



A collection of rights, privileges, obligations, and responsibilities delicately balanced over a period of time through conflict and conflict resolution.

How do these definitions of organizations relate to information systems technology?

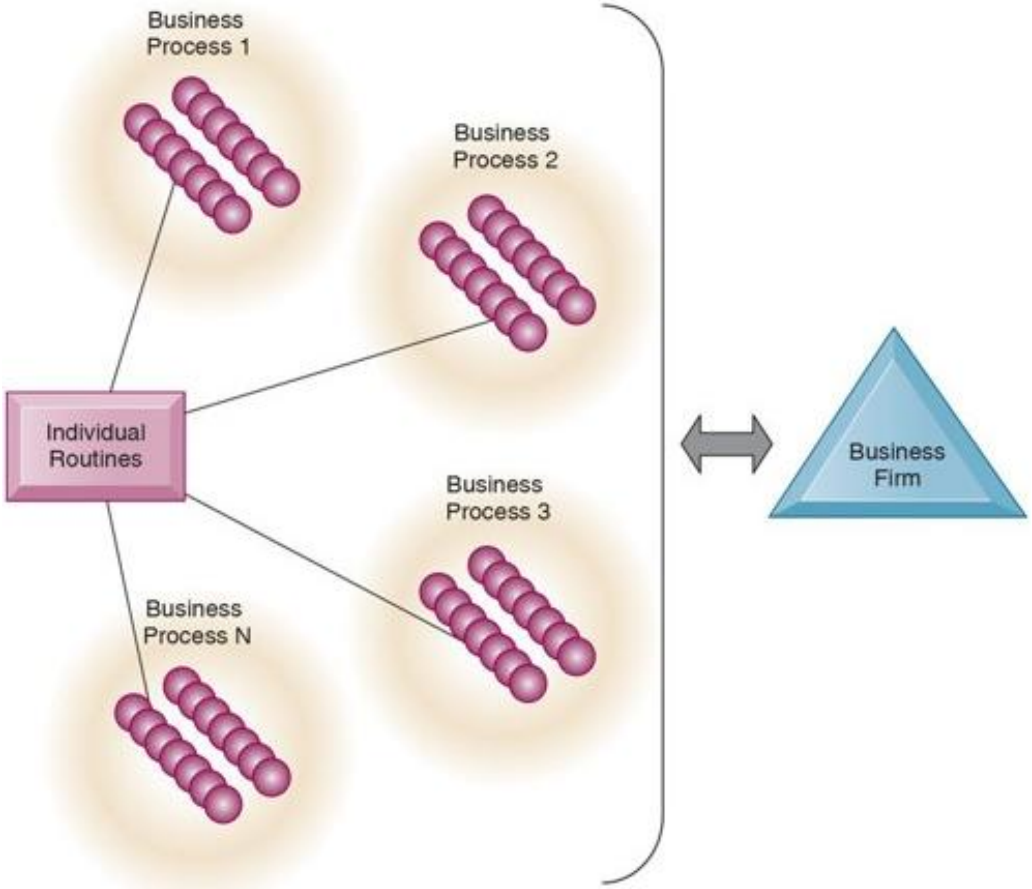
- Technical View
  - Focus on inputs → processes → outputs
  - **Org → IS:** how inputs and processes are structured, shaping system design for efficiency
  - **IS → Org:** automates processes and reallocates resources, improving productivity and performance.
- Behavioral View
  - Focus on group relationships, values, structures
  - **Org → IS:** culture, politics, and human behavior influence how systems are adopted and used
  - **IS → Org:** changes roles, power dynamics, and communication patterns, requiring adaptation and learning
- Integrative Insight
  - Successful IS implementation requires balancing technical efficiency with human and organizational realities.

Organizational Features Influencing Information Systems

- Routines (standard operating procedures, SOP)
  - Precise rules, procedures, and practices developed to cope with virtually all expected situations
- Business processes
  - Collection of routines

New IS require changes in routines and processes to improve organizational performance.

<u>Your examples?</u>



Organizational Features Influencing Information Systems

- Organizational Politics
  - Differences in interests, power, and perspectives among groups lead to competition and conflict over resources and decisions.
  - Impact on IS: Political **resistance** can **slow or block** system implementation and change; managing these dynamics is key to success.
- Organizational Environment
  - Organizations depend on and interact with external forces – customers, competitors, governments, and resources – in a continuous exchange.
  - Impact on IS: Information systems enable environmental scanning and strategic response, helping firms adapt to external changes and influence their surroundings.

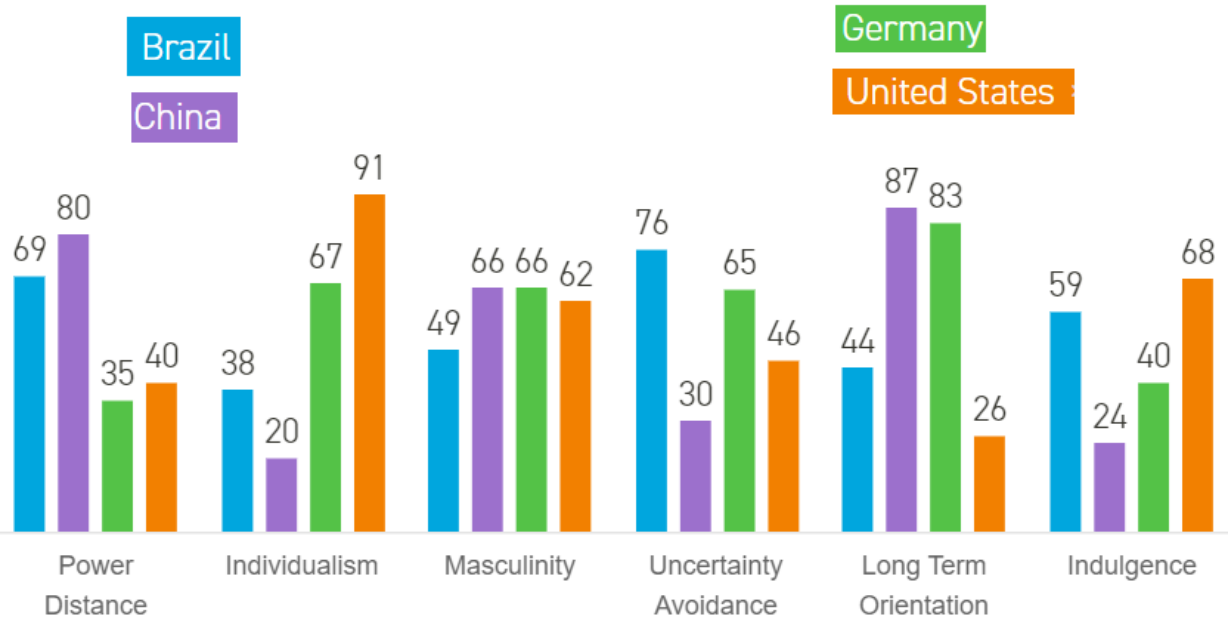
Organizational Features Influencing Information Systems

- Organizational Culture
  - A “collective programming of the mind” involving “shared values and beliefs.”
- Layers of Culture
  - Observable artifacts – most visible layer
    - E.g., dress, acronyms, awards, stories, rituals
  - Espoused values: explicitly stated preferred values
    - E.g., “We have a good work-life balance.”
  - Enacted values: reflected in actual behavior, sometimes inconsistent with espoused values
    - E.g., “We have a good work-life balance.” But, 996.
  - Assumptions – deepest layer – unobservable; taken for granted

Organizational Features Influencing Information Systems

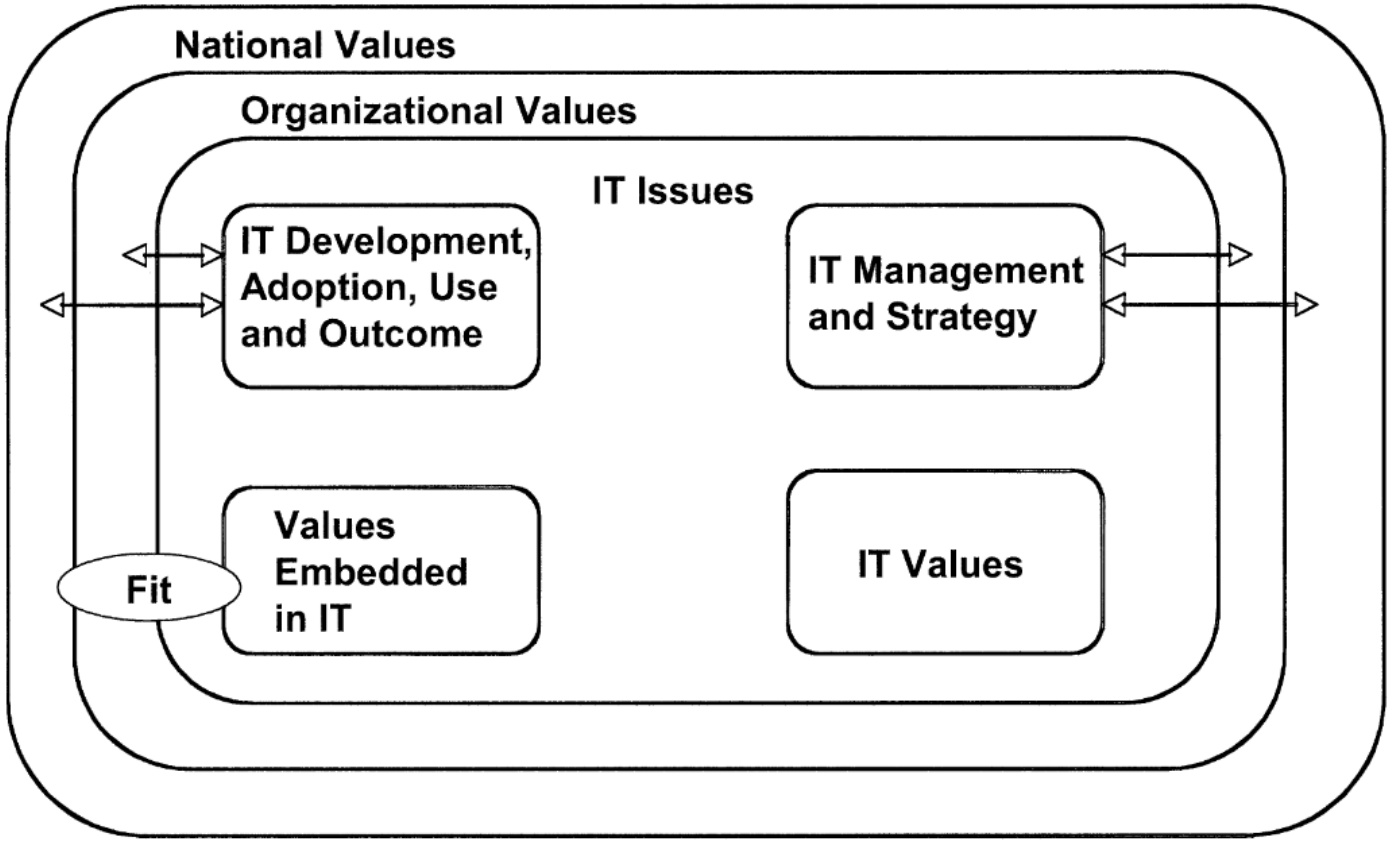
Hofstede's Cultural Dimensions Theory

- Dimensions of National Culture
  - Power distance index (PDI) 权力距离指数
  - Individualism vs. collectivism (IDV) 个人主义和集体主义
  - Uncertainty avoidance (UAI) 不确定性规避指数
  - Motivation towards Achievement and Success (MAS) 男性化和女性化
  - Long-term orientation vs. short-term orientation (LTO) 长期导向
  - Indulgence vs. restraint (IND) 放任与约束



Organizational Features Influencing Information Systems

How are IT and Culture Linked? (Leidner and Kayworth 2006)



Organizational Features Influencing Information Systems

Organizational Structure (Mintzberg, 1971)

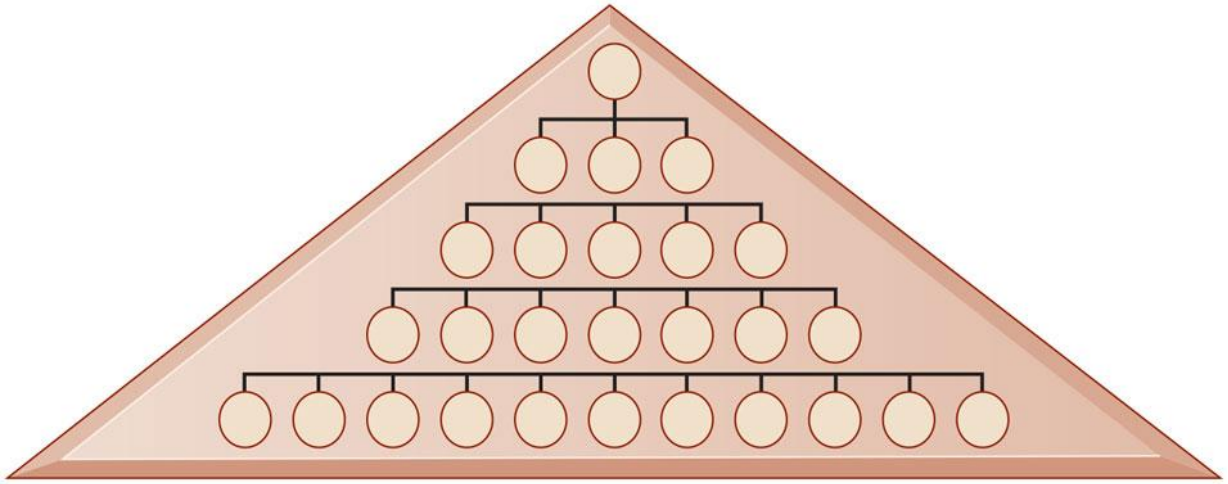
Organizational Type	Description	Examples
Entrepreneurial structure	Young, small firm in a fast-changing environment. It has a simple structure and is managed by an entrepreneur serving as its single chief executive officer.	Small start-up business
Machine bureaucracy	Large bureaucracy existing in a slowly changing environment, producing standard products. It is dominated by a centralized management team and centralized decision making.	Midsize manufacturing firm
Divisionalized bureaucracy	Combination of multiple machine bureaucracies, each producing a different product or service, all topped by one central headquarters.	*Fortune* 500 firms, such as General Motors
Professional bureaucracy	Knowledge-based organization where goods and services depend on the expertise and knowledge of professionals. Dominated by department heads with weak centralized authority.	Law firms, school systems, hospitals
Adhocracy	Task force organization that must respond to rapidly changing environments. Consists of large groups of specialists organized into short-lived multidisciplinary teams and has weak central management.	Consulting firms, such as the RAND Corporation

Information systems mirror organizational structure.

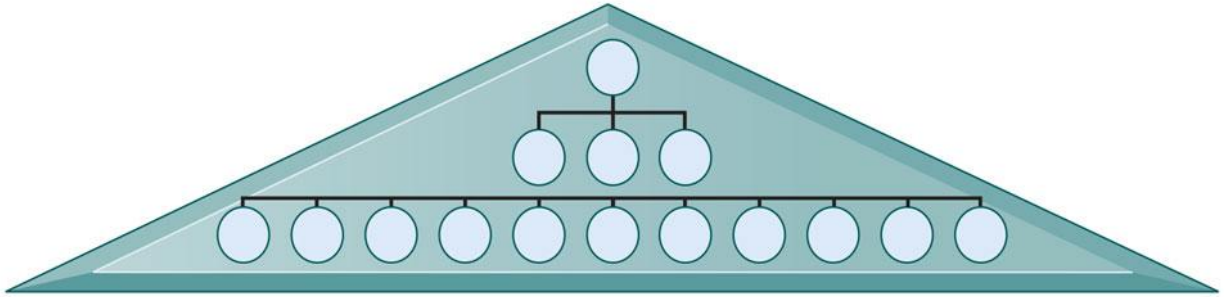
Impact of IS on Organizations – Economic

- IT as a Production Factor
  - Substitutes for labor and capital as technology costs decline.
- Labor & Capital Substitution
  - Automation reduces middle managers and clerical staff; firms invest more in IT instead of costly buildings or machinery.
- Transaction Cost Theory (Coase, Williamson)
  - IT reduces market participation costs (search, communication, monitoring), making outsourcing more efficient → firms shrink in size.
- Agency Theory (Jensen & Meckling)
  - IT lowers supervision and coordination costs, allowing managers to oversee more employees and flatten hierarchies.
- Overall Effect
  - Firms become leaner, more networked, and more productive, achieving higher revenue per employee.

Impact of IS on Organizations – Behavioral



A traditional hierarchical organization with many levels of management



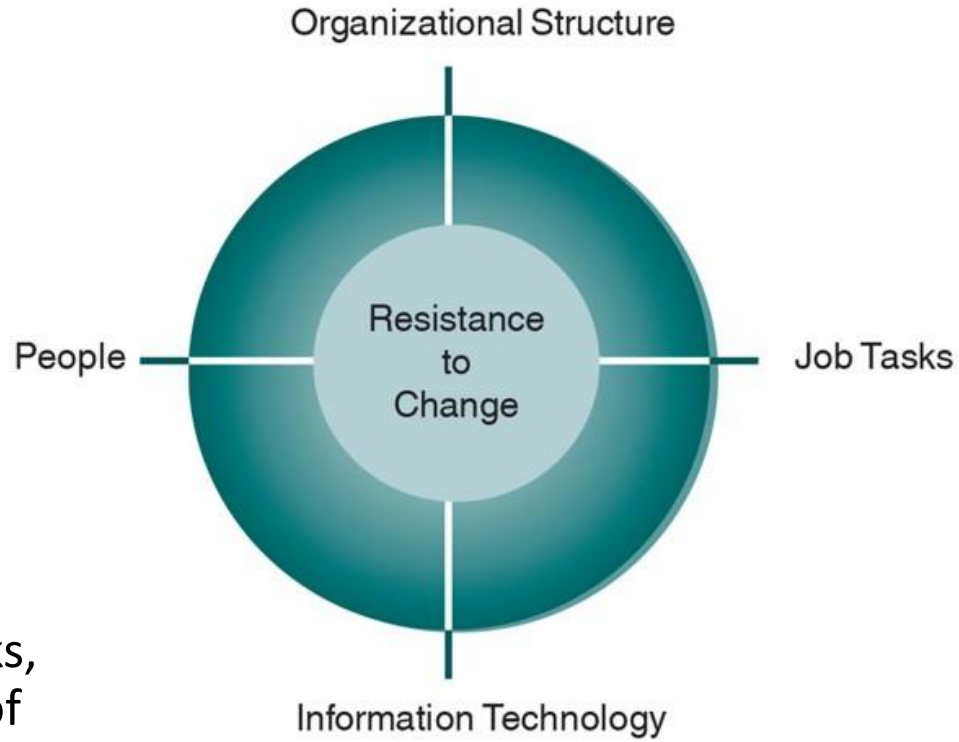
An organization that has been "flattened" by removing layers of management

IT flattens organizations

Information systems can reduce the number of levels in an organization by providing managers with information to supervise larger numbers of workers and by giving lower-level employees more decision-making authority.

Impact of IS on Organizations – Behavioral

- Why organizational resistance to change occurs?
  - IS reshapes access to information, roles, routines, and power—creating political tension and discomfort.
- Sources of resistance
  - Technology: perceived disruption or complexity.
  - Structure: misalignment with hierarchy or workflows.
  - People & Culture: habits, values, and fear of loss.
  - Tasks: changes in responsibilities or skills required.
- Key insight
  - Successful IS adoption requires adjusting technology, tasks, structure, and people together, often through a process of unfreezing → changing → refreezing.



Implications for IS Design and Implementation

To create real value, information systems must align with organizational context and dynamics. Key factors to consider:

- **Environment:** External forces and industry conditions.
- **Structure:** Hierarchy, specialization, and core processes.
- **Culture & Politics:** Shared values, norms, and power dynamics.
- **Leadership Style:** Decision-making and change orientation.
- **Stakeholders:** Interests and attitudes of system users.
- **Tasks & Decisions:** Business activities the system supports.

Summary

- IS and Organizations Are Interdependent
  - Information systems both shape and are shaped by organizational structure, culture, politics, and environment.
  - Effective systems must align with how the organization works and evolves.
- Organizational Features Matter
  - Routines, politics, culture, and structure influence IS design, adoption, and outcomes.
  - Managing resistance and change is essential for successful implementation.
- Impact of IS on Organizations
  - Economic: reduces transaction and coordination costs, flattens hierarchies.
  - Behavioral: empowers employees, redistributes information and power.
- Key Takeaway

  👉 To create real value, IS must be technically sound, organizationally aligned, and socially accepted.

Daily Feedback